Program : **Diploma in Cloud Computing and Big Data**	
Course Code : **4272**	Course Title: **Data Structures and Algorithms**
Semester : **4**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- Explain the fundamental concept of Data Structures and Algorithms.
- Emphasize the importance of data structures in designing and implementing efficient algorithms.

Course Prerequisites:

Topic	Course code	Course name	Semester
Sound knowledge in Python Programming	1008	Introduction to IT systems Lab	I
		Programming in Python	III
Control Structures, Modular Program Design, Arrays, Functions	2131	Problem Solving and Programming	II
	2139	Problem Solving and Programming Lab	II

Course Outcomes :

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic data structure concepts through python language	13	Applying
CO2	Use different Searching and Sorting Algorithms	13	Applying
CO3	Demonstrate linear data structures Stack, Queue and Linked Lists	16	Applying
CO4	Sketch nonlinear Data Structures such as Tree and graph	16	Applying

	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	1					
CO2	3	1					
CO3	3	1					
CO4	3	1					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Illustrate the basic data structure concepts through python language**		
M1.01	Describe the concept of algorithm and asymptotic Analysis	3	Understanding
M1.02	Demonstrate Recursive Algorithms	2	Applying
M1.03	Explain Data Structures and it's classifications	2	Understanding
M1.04	Illustrate Arrays in Python	6	Applying
Contents: **Algorithms:** Definition, Asymptotic Analysis of Algorithms- The "Big-Oh" Notation, Big Omega Notation, Recursion **Data Structures**: Definition, Classification- Linear Data Structures, Non-Linear Data Structures, Abstract Data Type(ADT) **Arrays**: Array based sequences,operations-Traverse,insert,delete,search,update -Two Dimensional Arrays in python			
CO2	**Use different Searching and Sorting Algorithms**		
M2.01	Demonstrate Linear Search algorithm	1	Applying
M2.02	Illustrate Binary Search algorithm	3	Applying
M2.03	Use Bubble Sort algorithm	1	Applying
M2.04	Apply Selection Sort algorithm	2	Applying
M2.05	Demonstrate Merge Sort algorithm	3	Applying

M2.06	Sketch Quick Sort algorithm	3	Applying

Contents:

Searching Algorithms: Linear Search, Binary Search.

Sorting Algorithms: Bubble Sort, Selection Sort, Quick Sort, Merge Sort

CO3	**Demonstrate linear data structures Stack, Queue and Linked Lists**		
M3.01	Demonstrate Array Based Stack ADT operations	1	Applying
M3.02	Use stack to Convert infix expression to postfix expression and evaluate postfix expression.	4	Applying
M3.03	Illustrate Array based Queue ADT	1	Applying
M3.04	Classify different type of Queues such as circular Queue and Double Ended Queue	3	Understanding
M3.05	Explain Linked list Concept	1	Understanding
M3.06	Demonstrate different types of linked list such as singly linked list,doubly linked list and circular linked list	3	Applying
M3.07	Implement stack ADT using singly Linked list	1	Applying
M3.08	Use a Singly Linked List to implement QueueADT	1	Applying

Contents:

Stacks: Introduction to Stack ADT - Array Representation of Stacks - Operations on a Stack - Applications of Stacks - Infix-to-Postfix Conversion - Evaluating Postfix Expressions.

Queues: Introduction to Queue ADT - Array Representation of Queues - Operations on a Queue - Types of Queues: Circular Queue – Dequeue.

Linked Lists: Singly Linked List - Representation in Memory, Operations on a Singly Linked List – Insertion, Deletion, Searching and Traversal.

Types of Linked List- Circularly Linked Lists, Doubly Linked Lists.

Linked List Representation of Stack and Queue

CO4	**Sketch nonlinear Data Structures like Tree and graph**		
M4.01	Explain the concepts in Graph data structure and its types	2	Understanding
M4.02	Apply the different representations of a Graph.	2	Applying

M4.03	Demonstrate Graph Traversal algorithms such as BFS and DFS.	3	Applying
M4.04	Describe Tree Data Structure and terminologies related to it	1	Understanding Applying
M4.05	Classify different types of Binary Trees	1	Understanding Applying
M4.06	Use different representations of Binary Tree	1	Applying
M4.07	Illustrate Different Tree Traversal Algorithms such as Inorder, Preorder and Postorder	2	Applying
M4.08	Implement Binary Search Tree operations	4	Applying

Contents:

Graphs: Graph Terminologies - Vertex, Edge, Adjacent vertices, Self-loop, Parallel edges, Isolated vertex, Degree of vertex, Pendant vertex, Subgraph, Paths and Cycles.

Types of Graphs - Directed, Undirected, Simple, Complete, Cyclic, Acyclic, Bipartite, Complete Bipartite, Connected, Disconnected and Regular.

Representation of Graphs - Set – Linked – Matrix

Graph Traversals-BFS, DFS

Trees: Definition - Basic Terminologies - Node, Parent, Child, Link, Root, Leaf, Level, Height of a tree and node, Depth of a tree and node, Degree of a tree and node, sibling, Ancestors, Path, Path Length.

Binary Tree- Types of Binary Trees: Full, Complete, Strict, Perfect.

Representations of a Binary Tree using Linked Lists

Tree Traversal Algorithms- Inorder, Preorder, Postorder.

Operations on a Binary Search Tree: Insertion –Traversal – Deletion– Count number of nodes – Height of a BST

Text / Reference

T/R	Author/Book Title
T1	Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, Data Structures and Algorithms in Python (An Indian Adaptation), Wiley; 1st edition (1 July 2021)
R1	Bradley N Miller Problem Solving with Algorithms and Data Structures Using Python, Franklin Beedle & Assoc; Second edition (1 January 2013)

Online Resources

Sl No	Website Link
1	https://www.tutorialspoint.com/python_data_structure/python_data_structure_introduction.htm
2	https://www.geeksforgeeks.org/python-data-structures/
3	https://www.edureka.co/blog/data-structures-in-python/
4	https://realpython.com/python-data-structures/
5	https://www.programiz.com/dsa